

Simulation Validation Interface Module (SVIM)

OOF™ Origin Open Foundation™

Independent Methodological Authority

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

Parent Standard: Simulation Evidence & Reliance Boundary Standard (SERBS)

Module: SVIM — Simulation Validation Interface Module

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Governed Space: Simulation-to-Validation Interface

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Canonical Definition

Simulation Validation Interface Module (SVIM) is the fifth Core Module of the Simulation Evidence & Reliance Boundary Standard (SERBS) within SIMULOS® — Simulation Governance Architecture.

SVIM governs the methodological transfer boundary through which qualified simulation-derived evidence leaves simulation-specific evidence governance and enters broader validation governance.

The Module ensures that Simulation Evidence transferred beyond SIMULOS® retains the identity, provenance, evidential purpose, qualification state, evidential weight, applicability, reliance boundary, uncertainty, dependencies, use limitations, and simulation basis necessary to preserve its legitimate meaning.

SVIM does not determine whether the transferred evidence is sufficient for validation, whether validation criteria have been satisfied, whether a Validation Determination may be issued, or whether operational reliance is justified.

Those questions belong to VALIDOS®.

Its task is narrower and architecturally critical:

to ensure that simulation-derived evidence crosses into broader validation governance without becoming detached from the conditions that define what that evidence legitimately means.

Core Question

How must simulation-derived evidence enter broader validation governance without losing the conditions that define its legitimate meaning?

Architectural Position

SVIM is the fifth and final module within SERBS.

It follows:

SEQM — Simulation Evidence Qualification Module → determines what simulation-derived information qualifies as evidence.

SEWM — Simulation Evidence Weight Module → determines how much simulation-specific evidential significance it may carry.

SRBM — Simulation Reliance Boundary Module → determines how far reliance may legitimately extend.

SUM — Simulation Use Limitation Module → determines what uses, claims, interpretations, or generalizations must remain constrained or prohibited.

SVIM then governs the controlled transfer of this complete simulation-specific evidential meaning into broader validation governance.

The sequence is:

Qualification → Weight → Reliance Boundary → Use Limitation → Validation Interface.

SVIM therefore represents the final simulation-evidence gateway before validation governance begins.

Internal Flow Position

Qualified Simulation Evidence Simulation Evidence Weight Simulation Reliance Boundary Simulation Use Limitations Simulation Evidence Provenance Material Uncertainty → Validation Transfer Purpose Identification → Evidence Package Assembly → Simulation Basis Preservation → Qualification Preservation → Weight Preservation → Boundary Preservation → Limitation Preservation → Uncertainty Preservation → Transfer Completeness Assessment → Validation Interface State → Simulation Validation Interface Package → VALIDOS® Validation Governance

Operational Space

SVIM governs the space between:

completed simulation-specific evidence governance

and

entry into broader validation governance.

This operational space includes: validation transfer purpose; evidence package identity; transferred evidence identity; simulation evidence provenance; originating Simulation Object linkage; simulation lifecycle traceability; evidential purpose preservation; qualification-state preservation; evidence-weight preservation; applicability preservation; reliance-boundary preservation; use-limitation preservation; uncertainty preservation; dependency disclosure; inference disclosure; correlation-basis preservation; negative-evidence preservation; contradictory-evidence preservation; transfer completeness; transfer versioning; evidence-package status; receiving validation context; validation-interface state; re-entry following evidence change.

Operational Scope

SVIM applies whenever simulation-derived evidence is transferred into a broader process intended to determine, support, challenge, reconsider, or inform: validation; system qualification; engineering acceptance; scientific validation; safety validation; mission validation; performance validation; deployment validation; operational readiness; regulatory validation; design validation; infrastructure validation; AI validation; autonomous-system validation; other governed validation processes.

It applies regardless of whether the transferred evidence originates from: one simulation; repeated simulations; Monte Carlo studies; scenario portfolios; large simulation campaigns; engineering simulations; scientific simulations; mission simulations; safety simulations; digital twins; physical simulations; hybrid simulations; AI simulations; autonomous-system simulations; future simulation technologies.

SVIM remains technology-neutral. Simulation Evidence Governance ≠ Validation Governance

SVIM enforces the architectural distinction:

Simulation Evidence Governance ≠ Validation Governance

SIMULOS® may establish: what evidence exists; what proposition it addresses; how it was produced; what it legitimately means; what simulation-specific weight it carries; where reliance upon it stops; what limitations constrain its use.

SIMULOS® must not then make the separate determination:

Is the complete validation basis sufficient?

That question belongs to VALIDOS®.

The interface exists precisely because these are different governance spaces.

Interface Principle

SVIM establishes:

A validation interface must transfer evidential meaning, not merely evidence

content.

A simulation result copied into a validation package without its governing conditions is not methodologically equivalent to the governed evidence object from which it originated.

Therefore:

Transferred Result ≠ Transferred Evidential Meaning

unless the material simulation-specific evidence conditions are preserved.

Validation Transfer Purpose

Every SVIM transfer should identify why simulation evidence is entering validation governance.

The transfer purpose may include: supporting a validation criterion; challenging a validation proposition; supplementing empirical evidence; supporting comparative validation; identifying unresolved uncertainty; demonstrating bounded simulated behavior; documenting failure conditions; contributing to a safety argument; supporting mission readiness evaluation; informing operational readiness; supporting a broader validation determination.

The transfer purpose must not silently expand the evidential purpose originally established under SEQM.

Simulation Validation Interface Package

SVIM establishes the Simulation Validation Interface Package as the governed transfer object between SIMULOS® and broader validation governance.

A package may contain one or multiple Simulation Evidence objects.

At minimum, where materially relevant, the package should preserve:

Evidence Identity → What simulation-derived evidence is being transferred?

Evidential Purpose → What proposition was the evidence qualified to address?

Simulation Basis → From which governed simulation conditions did the evidence originate?

Provenance → Which outputs, executions, scenarios, models, inputs, assumptions, and object configurations produced it?

Qualification State → Is the evidence Qualified, Conditionally Qualified, Suspended, or otherwise governed?

Evidence Weight → What simulation-specific evidential significance was assigned?

Applicability → Where does the evidence apply?

Reliance Boundary → How far may simulation-specific reliance legitimately

extend?

Use Limitations → What uses, interpretations, claims, or generalizations remain constrained?

Uncertainty → What uncertainty materially affects evidential meaning?

Correlation Basis → What simulation-to-reality relationship supports or constrains the evidence?

Dependencies → Which common models, assumptions, inputs, scenarios, or other dependencies affect independence?

Temporal Applicability → When does the evidence remain applicable? Evidence Content ≠ Evidence Context

SVIM establishes:

Evidence Content ≠ Evidence Context

A numerical value, chart, probability, trajectory, success rate, failure rate, confidence interval, comparison, or simulation conclusion may be correctly copied while its evidential meaning is incorrectly transferred.

For example:

99.97% simulated success

without: scenario domain; model basis; environmental representation; excluded conditions; failure treatment; uncertainty; object configuration; correlation basis;

does not preserve the original governed evidential meaning.

SVIM therefore treats contextual preservation as part of transfer governance.

Provenance Preservation

Simulation evidence entering validation must remain traceable to its originating simulation basis.

Where materially relevant, SVIM preserves:

Simulation Evidence → Simulation Output → Execution Record → Scenario → Input Basis → Model & Environment → Assumptions → Reality Representation → Purpose & Scope → Simulation Object

and, where applicable:

Simulation Evidence → Simulation-to-Reality Correlation Basis

SVIM does not replace general provenance or integrity governance.

It ensures that the simulation-specific provenance required to understand the evidence remains available at the interface.

Qualification Preservation

An evidence object must not acquire a stronger qualification state merely because it enters validation.

Therefore:

Conditionally Qualified Simulation Evidence must not become Unconditionally Qualified Validation Evidence

through transfer alone.

Likewise:

Qualification Suspended

must remain visible to the receiving validation process.

Transfer does not repair qualification deficiencies.

Evidence Weight Preservation

SVIM requires simulation-specific evidential weight to remain visible during transfer.

A validation process may subsequently evaluate the evidence differently within the complete validation basis.

But it must be able to identify the simulation-specific weight established under SEWM and the basis supporting that determination.

Therefore:

Validation Entry $\neq$ Automatic Evidence Weight Increase.

Reliance Boundary Preservation

The reliance boundary established under SRBM must remain attached to the evidence entering validation.

If evidence applies only to:

Object Version V3 + Scenario Family S7 + Environment E4 + Parameter Range P2

the validation process must receive that boundary.

The transfer must not silently convert it into evidence applicable to:

all object versions + all scenarios + all environments + unrestricted parameter ranges.

Use Limitation Preservation

All material SUM limitations must remain available at the validation interface.

These may include: prohibited generalization; prohibited extrapolation; configuration restriction; scenario restriction; environmental restriction; temporal restriction; uncertainty limitation; decision-class limitation; comparative-only use; prohibition against independent safety claims; requirement for additional empirical evidence.

SVIM establishes:

A limitation omitted at transfer is a limitation no longer capable of governing downstream interpretation.

Uncertainty Preservation

Simulation evidence must not become artificially more certain when transferred.

SVIM therefore requires material uncertainty to remain visible.

This may include uncertainty originating from: reality representation; assumptions; model structure; parameters; inputs; scenarios; execution; outputs; inference; simulation-to-reality correlation.

Validation transfer must not convert bounded uncertainty into apparent certainty.

Dependency Preservation

Large simulation campaigns may generate many evidence objects sharing common dependencies.

SVIM requires relevant dependencies to accompany evidence where they affect interpretation.

Examples include: common model; common assumption set; common input source; common scenario generator; common environment model; common transformation; common inference methodology; common correlation basis.

This allows validation governance to distinguish:

multiple evidence objects

from

multiple independent evidence sources.

Negative Evidence Preservation

SVIM must preserve materially relevant negative evidence.

A validation interface must not transfer only: successful runs; favorable outputs; confirming evidence; preferred scenarios.

Material: failures; anomalies; contradictions; divergence; instability; non-convergence; threshold violations; correlation failures;

must remain visible where relevant to the validation proposition.

Validation transfer must not become evidence selection.

Contradictory Evidence Preservation

Where simulation evidence objects materially disagree, the disagreement must cross the interface.

SVIM must not resolve contradiction by simply transferring the preferred evidence object.

The validation process may need to determine the significance of the contradiction.

SVIM's responsibility is to preserve it.

Inference Preservation

Where evidence is derived through inference rather than direct simulation observation, the validation interface must identify that inferential pathway.

The receiving validation process should be able to distinguish:

Direct Simulation Observation

from

Simulation-Derived Inference.

Where materially relevant, the transfer should preserve: originating outputs; inference methodology; transformations; assumptions; aggregation; inference uncertainty; inference limitations.

Correlation Basis Preservation

Where evidence makes or supports claims concerning reality, the relevant SRCS correlation basis should remain available.

This may include: reality reference; comparison conditions; correlation criteria; observed deviations; alignment basis; correlation uncertainty; applicability boundary.

SVIM does not determine whether correlation is sufficient.

It preserves the governed correlation information necessary for the receiving validation process.

Transfer Completeness

SVIM requires assessment of whether the evidence package is sufficiently complete for transfer.

Completeness is purpose-specific.

A package may be incomplete where it lacks materially necessary: evidence identity; evidential purpose; simulation provenance; qualification state; evidence-weight basis; applicability; reliance boundary; limitation; uncertainty; correlation basis; dependency disclosure; negative evidence; temporal applicability.

An incomplete package should not be represented as a complete simulation evidence transfer. Transfer Completeness $\neq$ Validation Sufficiency

SVIM establishes an important distinction:

Complete Transfer $\neq$ Sufficient Validation Basis

A perfectly constructed Simulation Validation Interface Package may still provide only a small part of the evidence needed for validation.

Conversely, a validation process may ultimately have sufficient evidence from multiple sources even where simulation evidence contributes only limited weight.

SVIM governs transfer quality.

VALIDOS® governs validation sufficiency.

Validation Interface States

SVIM may assign a governed interface state.

A minimum state model may include: 1. Interface Not Ready

Material evidence conditions required for transfer remain unresolved. 2. Conditionally Transferable

Evidence may enter validation only with explicit unresolved conditions or restrictions. 3. Transfer Ready

The simulation evidence package contains the material simulation-specific information required for governed transfer. 4. Transferred

The governed evidence package has entered the designated validation process. 5. Transfer Review Required

A material change has occurred that may affect evidence meaning after transfer. 6. Transfer Superseded

A newer governed evidence package replaces the previous transfer basis.

These states describe the simulation-validation interface.

They are not validation states. Validation Interface State $\neq$ Validation State

SVIM must not create a parallel validation determination.

Therefore:

Transfer Ready ≠ Validated

Transferred ≠ Validated

Conditionally Transferable ≠ Conditionally Validated

The interface state answers only:

Is the simulation evidence appropriately constituted for transfer into validation governance?

Versioned Evidence Transfer

Simulation evidence may evolve after: additional executions; model updates; new scenarios; revised assumptions; new reality data; new correlation analysis; evidence requalification; changed evidence weight; revised reliance boundaries; new use limitations.

SVIM should therefore support versioned Simulation Validation Interface Packages.

A new package should not silently overwrite the evidential basis of an earlier validation process.

Evidence Change After Transfer

Where evidence changes materially after transfer, SVIM should support notification or re-entry into the validation interface.

Possible triggers include: qualification suspension; evidence disqualification; degraded weight; changed reliance boundary; newly discovered limitation; contradictory evidence; new simulation failure; new correlation deviation; object change; reality change; resimulation.

The downstream validation basis may need reassessment.

SVIM identifies the interface condition.

VALIDOS® determines the validation consequence.

Interface Re-Entry

Evidence may cross the interface more than once.

The lifecycle may therefore become:

Simulation Evidence → Validation Interface → Validation Governance → New Simulation Evidence / Material Change → Interface Re-Entry → Validation Reassessment

This supports recursive governance rather than one-time evidence transfer.

Machine-Readable Validation Interface

SVIM should support machine-readable evidence transfer where simulation and validation pipelines are automated.

Machine-readable interface fields may include: evidence ID; simulation object ID; object version; evidence version; evidential purpose; qualification state; evidence-weight state; applicability; reliance boundary; use limitations; uncertainty; dependency references; scenario references; execution references; output references; correlation references; temporal applicability; transfer state; supersession relationship.

This allows automated validation systems to ingest simulation evidence without stripping away governance context.

Human-Readable Validation Interface

Machine-readable transfer does not eliminate the need for understandable human interpretation.

SVIM therefore supports human-readable statements such as:

What the evidence shows.

What it does not show.

Under which conditions it applies.

What uncertainty remains.

How strongly the simulation evidence supports the proposition.

Where reliance must stop.

What additional validation may be necessary.

SVIM Boundary with SEQM

SEQM qualifies simulation-derived information as evidence.

SVIM transfers that qualified evidence into validation governance.

SVIM must not independently requalify evidence except where transfer review identifies a condition requiring return to SEQM.

SVIM Boundary with SEWM

SEWM determines simulation-specific evidential weight.

SVIM preserves that weight and its basis during transfer.

It does not determine validation-level evidence weight within the total validation basis.

SVIM Boundary with SRBM

SRBM establishes the simulation reliance boundary.

SVIM preserves that boundary when evidence enters validation.

It does not broaden the boundary.

SVIM Boundary with SUM

SUM establishes evidence-use limitations.

SVIM ensures those limitations survive transfer.

Therefore:

SUM defines the limitation.

SVIM preserves the limitation across the validation boundary.

SVIM Boundary with VALIDOS®

This is the primary architectural boundary governed by SVIM.

SVIM / SIMULOS®: packages simulation-derived evidence; preserves simulation-specific meaning; preserves qualification; preserves evidence weight; preserves applicability; preserves uncertainty; preserves reliance boundaries; preserves limitations; preserves provenance; transfers the evidence.

VALIDOS®: identifies the Validation Object; establishes Validation Purpose & Scope; applies validation criteria; governs validation methodology; evaluates validation evidence fitness; evaluates source reliability; executes validation; issues Validation Determinations; governs Validation States; governs reliance and revalidation.

The boundary is:

SIMULOS® prepares and transfers simulation-derived evidence.

VALIDOS® determines what that evidence means within the complete validation basis.

SVIM Boundary with INTEGROS®

SVIM does not establish general evidence integrity.

INTEGROS® may govern integrity properties of the evidence package, records, transformations, and transfer mechanisms.

SVIM governs which simulation-specific evidential conditions must survive the transfer.

SVIM Boundary with ADIT®

ADIT® may audit: whether the transfer occurred; what package was transferred; whether records were preserved; whether evidence conditions were omitted or altered; whether transfer history is reconstructable.

SVIM defines the simulation-specific transfer structure.

It does not own audit governance.

SVIM Boundary with ORA™

Operational Reality information may become relevant to validation.

SVIM does not determine Operational Reality.

It preserves the simulation-specific relationship to reality established through SRCS where that relationship is material to transferred evidence.

SVIM Boundary with AGA™

AGA™ governs responsibility, authority, accountability, and attribution for validation-related decisions.

SVIM does not determine who may approve validation, accept evidence, authorize deployment, or assume operational risk. SVIM Boundary with AIG® / ASGA™

Where simulation evidence concerns AI or autonomous systems, SVIM may transfer that evidence into validation governance.

It does not govern the behavior, authority, autonomy, permissions, constraints, or deployment of the actual AI or autonomous system.

Minimum Implementation Framework

A minimum SVIM implementation should establish: 1. Transfer Purpose Identification

Define why the Simulation Evidence is entering broader validation governance.

2. Evidence Package Identification

Identify the precise evidence object or evidence set being transferred. 3. Simulation Basis Preservation

Preserve traceability to the governed simulation lifecycle that produced the evidence. 4. Qualification Preservation

Transfer the current evidence qualification state and qualification basis. 5. Evidence Weight Preservation

Transfer the simulation-specific evidence weight and material factors supporting or constraining it. 6. Reliance Boundary Preservation

Transfer the maximum simulation-specific reliance boundary. 7. Use Limitation Preservation

Transfer material restrictions, exclusions, caveats, unsupported interpretations, and prohibited uses. 8. Uncertainty and Dependency Preservation

Transfer material uncertainty and evidence dependencies relevant to interpretation. 9. Transfer Completeness Assessment

Determine whether the package contains the simulation-specific information necessary for governed validation entry. 10. Interface State Assignment

Assign and document the current Validation Interface State. 11. Transfer Record

Record the evidence package version, receiving validation context, transfer state, and material interface conditions. 12. Change and Re-Entry Control

Define conditions requiring updated transfer, supersession, or validation-interface re-entry.

Governance Outputs

SVIM may produce: Simulation Validation Transfer Purpose Statement Simulation Validation Evidence Package Simulation Evidence Transfer Manifest Simulation Basis Transfer Record Simulation Evidence Qualification Transfer Record Simulation Evidence Weight Transfer Record Simulation Reliance Boundary Transfer Statement Simulation Use Limitation Transfer Record Simulation Evidence Uncertainty Transfer Record Simulation Evidence Dependency Disclosure Simulation Negative Evidence Disclosure Simulation Correlation Basis Transfer Record Simulation Validation Interface State Simulation Validation Interface Package Simulation Evidence Transfer Record Simulation Validation Interface Re-Entry Record

Together these outputs form the governed interface between:

SIMULOS® Simulation Evidence Governance

and

VALIDOS® Validation Governance.

Enterprise Application

SVIM solves a major organizational problem:

simulation teams and validation teams frequently work with the same results but not necessarily with the same evidential meaning.

A simulation team may know that a result: applies only to one configuration; depends upon specific assumptions; covers a defined scenario family; excludes particular environmental conditions; contains significant uncertainty; derives from dependent simulations; has limited reality correlation; supports comparison rather than absolute prediction.

Yet the downstream validation package may contain only:

the final number, graph, success percentage, or conclusion.

At that point the result has crossed organizational boundaries while its governance context has not.

SVIM prevents this loss.

For a space mission, a simulation package transferred into mission validation

can retain the exact mission configuration, represented environment, scenario coverage, assumptions, correlation basis, uncertainty, failures, and permitted evidential meaning.

For an autonomous vehicle or robot, millions of simulated scenarios can enter validation together with their shared model dependencies, unrepresented conditions, sim-to-real limitations, failure distributions, scenario boundaries, and evidence-weight assessment.

For engineering systems, simulation-derived structural, thermal, aerodynamic, mechanical, electrical, or reliability evidence can enter validation without becoming detached from the model, parameter range, configuration, assumptions, and conditions that produced it.

SVIM therefore creates a governed handover protocol for evidential meaning .

It enables organizations to use simulation aggressively while preventing the organizational transfer process from silently converting bounded simulation evidence into apparently unrestricted validation evidence.

Foundational Principles

Simulation evidence governance and validation governance are distinct governance spaces.

A validation interface must transfer evidential meaning, not merely evidence content.

Evidence content without evidence context is an incomplete evidential transfer.

Conditional qualification must remain conditional after transfer.

Simulation-specific evidence weight must not increase merely because evidence enters validation.

Reliance boundaries must survive validation transfer.

Use limitations must survive validation transfer.

Material uncertainty must survive validation transfer.

Evidence dependencies must survive validation transfer.

Negative and contradictory evidence must not disappear at the validation boundary.

Transfer completeness is not validation sufficiency.

Transfer Ready does not mean Validated.

A simulation result must not acquire greater certainty, generality, applicability, independence, or authority merely by crossing from simulation governance into validation governance.

SIMULOS® governs the legitimate simulation-specific evidential meaning carried to the boundary; VALIDOS® governs what the complete validation basis can legitimately determine beyond that boundary.

Canonical Closing Statement

Simulation Validation Interface Module establishes the final evidential transfer layer of the Simulation Evidence & Reliance Boundary Standard by governing how qualified simulation-derived evidence crosses from SIMULOS® simulation governance into broader validation governance without losing the conditions that define its legitimate meaning. Through preservation of evidence identity, evidential purpose, simulation basis, provenance, qualification state, evidential weight, applicability, reliance boundaries, use limitations, uncertainty, dependencies, inference pathways, correlation basis, negative evidence, contradictory evidence, temporal applicability, transfer state, versioning, and interface re-entry, SVIM prevents bounded simulation evidence from becoming artificially broader, stronger, more certain, or more authoritative through organizational, methodological, or automated transfer. By separating transfer completeness from validation sufficiency and Validation Interface States from Validation States, SVIM preserves the architectural boundary between SIMULOS® and VALIDOS® while enabling simulation-derived evidence to enter validation as a fully contextualized and governable evidential object. The simulation lifecycle therefore reaches the validation boundary not with an isolated result, but with a reconstructable statement of what the evidence is, where it came from, what it supports, how strongly it supports it, where reliance stops, what uncertainty remains, and what the evidence must never be used to claim.

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Canonical Source

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